

3.5: Dividing Polynomials

Example. Compute $178 \div 3$ using long division. Identify the dividend, divisor, quotient, and remainder.

$$\begin{array}{r}
 59 \\
 3 \overline{) 178} \\
 \underline{-(15)} \\
 28 \\
 \underline{-27} \\
 1
 \end{array}$$

$$178 = 3 \cdot 59 + 1$$

$\uparrow \qquad \qquad \uparrow \qquad \qquad \uparrow \qquad \qquad \uparrow$
 Dividend = (divisor $\cdot$ quotient) + remainder

$$\frac{178}{3} = 59 + \frac{1}{3} = 59 R 1$$

Definition. (The Division Algorithm)

The **Division Algorithm** states that, given a polynomial dividend $f(x)$ and a non-zero polynomial divisor $d(x)$ where the degree of $d(x)$ is less than or equal to the degree of $f(x)$, there exist unique polynomials $q(x)$ and $r(x)$ such that

$$f(x) = d(x)q(x) + r(x)$$

where $q(x)$ is the quotient and $r(x)$ is the remainder. The remainder is either equal to zero, or has degree strictly less than $d(x)$.

If $r(x) = 0$, then $d(x)$ divides evenly into $f(x)$. This means that, in this case, both $d(x)$ and $q(x)$ are factors of $f(x)$.

Example. Use polynomial long division on the following:

$$\frac{5x^2 + 3x - 2}{x + 1} = 5x - 2 + \frac{0}{x+1}$$

$$\begin{array}{r} 5x - 2 \\ x+1 \overline{) 5x^2 + 3x - 2} \\ \underline{-(5x^2 + 5x)} \\ -2x - 2 \\ \underline{-(-2x - 2)} \\ 0 \end{array}$$

$$\frac{2x^2 + 5x - 7}{3x - 2} = \frac{2}{3}x + \frac{19}{9} + \frac{-25/9}{3x-2}$$

$$\begin{array}{r} \frac{2}{3}x + \frac{19}{9} \\ 3x-2 \overline{) 2x^2 + 5x - 7} \\ \underline{-(2x^2 - \frac{4}{3}x)} \\ \frac{19}{3}x - 7 \\ \underline{-(\frac{19}{3}x - \frac{38}{9})} \\ -\frac{25}{9} \end{array}$$

Definition. (Synthetic Division)

Synthetic division is a shortcut that can be used when the divisor is of the form $(x - k)$. In **synthetic division**, only the coefficients are used in the division process.

Given two polynomials, use synthetic division to divide:

1. Write k for the divisor.
2. Write the coefficients of the dividend.
3. Bring the lead coefficient down.
4. Multiply the lead coefficient by k . Write the product in the next column.
5. Add the terms of the second column.
6. Multiply the result by k . Write the product in the next column.
7. Repeat steps 5. and 6. for the remaining columns.
8. Use the bottom numbers to write the quotient. The number in the last column is the remainder.

Example. Use synthetic division on the following:

$$\frac{5x^2 + 3x - 2}{x + 1} = 5x - 2$$

$$\begin{array}{r|rrr} -1 & 5 & 3 & -2 \\ & & -5 & 2 \\ \hline & 5 & -2 & 0 \end{array}$$

$$\begin{aligned} \frac{2x^2 + 5x - 7}{3x - 2} &= \frac{2x^2 + 5x - 7}{3(x - \frac{2}{3})} \\ &= \frac{2x + 19/3}{3} + \frac{-25/9}{3(x - 2)} \end{aligned}$$

$$\begin{array}{r|rrr} \frac{2}{3} & 2 & 5 & -7 \\ & & \frac{4}{3} & \frac{38}{9} \\ \hline & 2 & \frac{19}{3} & -\frac{25}{9} \end{array}$$

Example. Use synthetic division on the following:

$$\frac{5x^2 - 3x - 36}{x - 3} = 5x + 12 + \frac{0}{x - 3}$$

$$\begin{array}{r|rrrr} 3 & 5 & -3 & -36 & \\ & & 15 & 36 & \\ \hline & 5 & 12 & 0 & \end{array}$$

$$\frac{4x^3 + 10x^2 - 6x - 20}{x + 2} = 4x^2 + 2x - 10$$

$$\begin{array}{r|rrrrr} -2 & 4 & 10 & -6 & -20 & \\ & & -8 & -4 & 20 & \\ \hline & 4 & 2 & -10 & 0 & \end{array}$$

$$\frac{-9x^4 + 10x^3 + 7x^2 - 6}{x - 1}$$

$$= -9x^3 + x^2 + 8x + 8 + \frac{2}{x - 1}$$

$$\begin{array}{r|rrrrrr} 1 & -9 & 10 & 7 & 0 & -6 & \\ & & -9 & 1 & 8 & 8 & \\ \hline & -9 & 1 & 8 & 8 & 2 & \end{array}$$